

Kanak Singh

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TECHNICAL SKILLS

Programming Languages: Java, C++, SQL, JavaScript

Web Technologies: HTML, CSS

Frameworks & Libraries: Spring Boot, Spring Security, Spring Data MongoDB, Spring Data Redis, Spring Kafka, Spring Mail, Lombok

Databases: MySQL, PostgreSQL, MongoDB Atlas, Redis

Tools & Technologies: Docker, Maven, Git, GitHub, Postman, IntelliJ IDEA, Swagger/OpenAPI, SonarQube

Core Concepts: RESTful APIs, JWT Authentication, OAuth2, BCrypt Password Hashing, Caching, Message Queuing, Scheduled Jobs, Sentiment Analysis

CS Fundamentals: Object-Oriented Programming (OOPS), Operating Systems, Database Management Systems (DBMS), Computer Networks, Data Structures & Algorithms (300+ problems solved)

PROJECTS

Journal App | *Spring Boot, MongoDB, Redis, Kafka, JWT, OAuth2, Swagger*

2025 - 2026

– 🌐 GitHub | 🌐 Live Demo

- Engineered **15+ REST APIs** with **JWT authentication** and **role-based authorization** (USER/ADMIN), securing 100% of protected endpoints
- Integrated **Google OAuth2** alongside custom credential-based auth, supporting **2 login methods** for seamless user onboarding
- Reduced external API response time by **60%** using **Redis caching** on Weather API calls with **300-second TTL**, eliminating redundant HTTP requests
- Built automated **weekly sentiment analysis engine** via `@Scheduled` cron jobs, processing **4 mood categories** (HAPPY/SAD/ANGRY/ANXIOUS) across all opted-in users every Sunday
- Designed **Apache Kafka** async messaging pipeline for email delivery with **direct fallback**, achieving **0% message loss** on sentiment reports
- Cut repeated DB calls by **80%** using in-memory **AppCache** loaded at startup via `@PostConstruct`, reducing MongoDB load significantly
- Configured **MongoDB transactions** with `@Transactional` ensuring **100% atomicity** across journal entry save and user update operations
- Deployed on **Railway** with **8 environment variables** securing all credentials — zero hardcoded secrets in codebase

Text Analyzer | *Spring Boot, MongoDB, REST API, Validation, Actuator, HTML/CSS/JS*

2026

– 🌐 GitHub | 🌐 Live Demo

- Built a **full-stack sentiment analysis app** with Spring Boot backend and vanilla JS frontend, serving **3 REST endpoints** from a single deployable JAR
- Designed custom **sentiment scoring algorithm** processing **70+ keyword dictionary** to classify text as POSITIVE, NEGATIVE, or NEUTRAL with **4 statistical metrics** (word, character, sentence count, sentiment score)
- Implemented **Bean Validation** with `@Valid` and `@NotBlank` enforcing **5000-character input limit** and `@RestControllerAdvice` handling **3 exception types** globally
- Integrated **Spring Boot Actuator** exposing **2 monitoring endpoints** (health, info) for real-time application observability
- Persisted **100% of analysis results** in MongoDB with **3 retrieval endpoints** — all results, by sentiment, and health check
- Deployed on **Railway** with **256MB memory optimization** via JVM flags, reducing out-of-memory errors by **100%**

ACHIEVEMENTS

- Solved **300+ DSA problems** on LeetCode and GeeksforGeeks.
- Participated in the **All India Hackathon** and developed a project-based solution as part of a team.
- Developed **2 production-grade backend projects** using Spring Boot, JWT Authentication, Redis Caching, and Kafka — both deployed and live.

EDUCATION

Guru Gobind Singh Indraprastha University(GGSIPU)

Bachelor of Technology | BPIT

Delhi, India

2023 – 2027